

1.3 Addition and Subtract

49

الشرط:
لازم نفس
الحجم

$$\text{Ex. } \begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix}_{2 \times 2} + \begin{bmatrix} 2 & -1 \\ 4 & 1 \end{bmatrix}_{2 \times 2} = \begin{bmatrix} 3 & 1 \\ 3 & 1 \end{bmatrix}_{2 \times 2}$$

$$\text{Ex. } \begin{bmatrix} 1 & -1 \\ 5 & 4 \\ 1 & 0 \end{bmatrix}_{3 \times 2} - \begin{bmatrix} 2 & 1 \\ 0 & 0 \\ 4 & 1 \end{bmatrix}_{3 \times 2} = \begin{bmatrix} -1 & -2 \\ 5 & 4 \\ -3 & -1 \end{bmatrix}_{3 \times 2}$$

$$\text{Ex. } \begin{bmatrix} 1 & 2 \\ 0 & 4 \end{bmatrix}_{2 \times 2} + \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}_{3 \times 1} \text{ is undefined}$$

different size

Properties

Eng. Al-sfeer
99496600

$$(1) A + B = B + A \quad \text{الجمع ابدالي}$$

$$(2) c \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} ca_{11} & ca_{12} \\ ca_{21} & ca_{22} \end{bmatrix}$$

$$\text{Ex. } 4 \begin{bmatrix} 1 & 0 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ -8 & 16 \end{bmatrix}$$

Special Matrices

* Identity Matrix (I_n) المصفوفة
الوحدة

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

* Zero Matrix (O_n) المصفوفة الصفرية

$$O_2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad O_3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad O_4 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

* Square Matrix المصفوفة المربعة

of rows = # of columns

Multiplication of A and B : ضرب مصفوفتين

Let $A : m \times p$ and $B : p \times n$

$$\therefore AB = (\text{row } A)(\text{col. } B) : m \times n$$

شرط ضرب مصفوفتين

أن يكون عدد الأعمدة في الأولى يساوي عدد الأسطر في الثانية.

طريقة الضرب

نضرب كل سطر في الأولى بكل عمود في الثانية ، مع مراعاة أن ضرب سطر في عمود ينتج عنصر واحد فقط .

Note:

$$(1) \begin{matrix} A & \times & B & = & C \\ 2 \times 3 & & 3 \times 2 & & 2 \times 2 \end{matrix}$$

$$(2) \begin{matrix} X & \times & Y & = & Z \\ 1 \times 2 & & 2 \times 4 & & 1 \times 4 \end{matrix}$$

$$(3) \begin{matrix} A & \times & B & = & Undefined \\ 2 \times 3 & & 2 \times 3 \end{matrix}$$

لأن عدد الأعمدة في الأولى لا يساوي عدد الأسطر في الثانية .

$$(4) AB \neq BA \quad \text{الضرب في المصفوفات غير إبدالي}$$

Sometimes $AB = BA$

~~مثال 1: إذا كانت A = (1 3) و B = (2 5) فإن AB = (2 16) و BA = (3 17) أي AB ≠ BA~~

Let $A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 2 \end{bmatrix}_{2 \times 3}$ $B = \begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix}_{2 \times 2}$ $C = \begin{bmatrix} 1 & -1 \\ 2 & 3 \\ 0 & 1 \end{bmatrix}_{3 \times 2}$

(a) Find $A(B+C)$

(b) BAC

Text Book

(a) $(B+C)$ is undefined because
size of $B \neq$ size of C

$\therefore A(B+C)$ is undefined

(b) $BA = \begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix}_{2 \times (2)} \begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 2 \end{bmatrix}_{(2) \times 3} = \begin{bmatrix} 6 & 1 & 13 \\ 0 & -1 & 2 \end{bmatrix}_{2 \times 3}$

$BAC = \begin{bmatrix} 6 & 1 & 13 \\ 0 & -1 & 2 \end{bmatrix}_{2 \times (3)} \begin{bmatrix} 1 & -1 \\ 2 & 3 \\ 0 & 1 \end{bmatrix}_{(3) \times 2} = \begin{bmatrix} 8 & 10 \\ -2 & -1 \end{bmatrix}_{2 \times 2}$

H.W. Find CA

Ans. $\begin{bmatrix} 2 & 2 & 1 \\ 4 & -1 & 12 \\ 0 & -1 & 2 \end{bmatrix}$

(5+5+5+5 pts) Let

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix}, \quad \text{and} \quad C = \begin{bmatrix} 1 & -1 \\ 2 & 3 \\ 0 & 1 \end{bmatrix}.$$

If possible, compute:

i) $A(B+C)$;

ii) $B(A+C^T)$;

iii) $(A^T+C)(C+A)$;

iv) BAC .

(i) $B+C$ is undefined $\left[\text{size}(B) \neq \text{size}(C) \right]$
 $2 \times 2 \quad 3 \times 2$

$\therefore A(B+C)$ is undefined

$$(ii) \quad A+C^T = \begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 3 & 3 \\ -1 & 2 & 3 \end{bmatrix}$$

$$B(A+C^T) = \begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 3 & 3 \\ -1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 7 & 13 & 15 \\ -1 & 2 & 3 \end{bmatrix}$$

$2 \times 2 \qquad 2 \times 3 \qquad 2 \times 3$

(iii) $C+A$ is undefined $\left[\text{size}(C) \neq \text{size}(A) \right]$
 $3 \times 2 \quad 2 \times 3$

$\therefore (A^T+C)(C+A)$ is undefined

$$(iv) \quad BA = \begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \\ 0 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 6 & 1 & 13 \\ 0 & -1 & 2 \end{bmatrix}$$

$2 \times 2 \qquad 2 \times 3 \qquad 2 \times 3$

$$(BA)C = \begin{bmatrix} 6 & 1 & 13 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & 3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 10 \\ -2 & -1 \end{bmatrix}$$

$2 \times 3 \qquad 3 \times 2 \qquad 2 \times 2$

Past Exam

Find A and B if

54

حل معادلتين في مجهولين

$$A - 2B = I_2 \quad \& \quad 2A - 3B = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}.$$

$$A = I_2 + 2B \rightarrow A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + 2B \rightarrow \textcircled{1}$$

$$\therefore 2A - 3B = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} \text{ ; from } \textcircled{1}$$

$$2 \left(\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + 2B \right) - 3B = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} + 4B - 3B = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}$$

Substitute in (1)

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + 2 \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 2 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$$

Quiz

Find A and B such that

$$A + 3B = \begin{bmatrix} 16 & 20 \\ 24 & 28 \end{bmatrix} \text{ and } 3A - 2B = \begin{bmatrix} -7 & -6 \\ -5 & -4 \end{bmatrix}$$

Past Exam

Ex: Find all values of K , if any, that satisfy the equation

Text Book

$$[K \ 1 \ 1] \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 2 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} K \\ 1 \\ 1 \end{bmatrix} = 0$$

1×3

3×3

3×1

$$[K+1 \quad K+2 \quad -1] \begin{bmatrix} K \\ 1 \\ 1 \end{bmatrix} = 0$$

1×3 3×1

$$[K(K+1) + (K+2)(1) + (-1)(1)] = 0$$

$$\therefore K^2 + K + K + 2 - 1 = 0$$

$$K^2 + 2K + 1 = 0$$

$$(K + 1)(K + 1) = 0$$

$$K = -1$$

* Powers in Matrices

الشرط: لازم المصفوفة Square

If A, B be $n \times n$ matrices قوانين الأسس

(1) $A^0 = I$; $A^2 = A A$; $A^3 = A^2 A$

(2) $A^p A^q = A^{p+q}$

$A^2 A^5 = A^7$

(3) $(A^p)^q = A^{pq}$

$(A^3)^2 = A^6$

(4) صحيح إيجابي

صحيح غير إيجابي

$(AB)^2 = A^2 B^2$

$(AB)^2 = (AB)(AB)$
 $= A(BA)B$

$(A+B)^2 = A^2 + 2AB + B^2$

$(A+B)^2 = (A+B)(A+B)$
 $= A^2 + AB + BA + B^2$

$(AB)^3 = A^3 B^3$

$(AB)^3 = (AB)(AB)(AB)$

(5) $(I_n)^k = I$

ملاحظات :-

$$1) A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$A^2 = AA \quad \checkmark$$

هذا الحل صحيح

$$A^2 = \begin{bmatrix} 1^2 & 2^2 \\ 3^2 & 4^2 \end{bmatrix} \quad \times$$

هذا الحل غير صحيح

$$2) \text{ Let } A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 1 \end{bmatrix}$$

A^2, A^3, A^4 are undefined

because A is NOT a square !!

3) خطوات الحل لمطلب A أس عدد عالي

1- نكتب A^2, A^3, A^4, A^5

2- نستخرج علاقة رياضية بينهم

< شوف الماتة القادمة الجز b >

58

Example : Let $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

Past Exam

(a) Find A^2 , A^3 and A^0 .

(b) Guess a formula for A^n for any integer n

② $A^2 = A \cdot A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$

③ $A^3 = A^2 \cdot A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$

④ $A^0 = I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $\rightarrow$ 2×2 : A $\rightarrow$ $n \times n$
 $A^0 = I_2$ $\rightarrow$ $n \times n$

④ $A^4 = A^2 \cdot A^2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix}$

⑤ $A^5 = A^4 \cdot A = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 5 \\ 0 & 1 \end{bmatrix}$

⑥ $A^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$ $\rightarrow$ 2016 $A = \begin{bmatrix} 1 & 2016 \\ 0 & 1 \end{bmatrix}$

Let $A = \begin{bmatrix} x & 1 \\ y & 2 \end{bmatrix}$. Find real numbers x, y such that $A^2 = 0$.

$$A^2 = AA = 0$$

$$\begin{bmatrix} x & 1 \\ y & 2 \end{bmatrix} \begin{bmatrix} x & 1 \\ y & 2 \end{bmatrix} = 0$$

$$\begin{bmatrix} x^2 + y & x + 2 \\ xy + 2y & y + 4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$x + 2 = 0 \rightarrow \boxed{x = -2}$$

$$y + 4 = 0 \rightarrow \boxed{y = -4}$$

check $\begin{bmatrix} 4 - 4 & 0 \\ 8 - 8 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \underline{\underline{OK}}$

Find a, b , and x such that

$$\begin{bmatrix} -1 \\ 1 \\ x \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + b \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$$

Text Book

Past Exam

60. انشاء x و y أعداد
رئیس مصفوفات

Let $A = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$, $a \neq 0$. Find the real numbers

x and y such that $A^4 + xA^2 + yI_2 = O_2$

$$A^2 = A \cdot A = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2a \\ 0 & 1 \end{bmatrix}$$

$$A^4 = A^2 \cdot A^2 = \begin{bmatrix} 1 & 2a \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2a \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 4a \\ 0 & 1 \end{bmatrix}$$

$$A^4 + xA^2 + yI_2 = O_2$$

$$\begin{bmatrix} 1 & 4a \\ 0 & 1 \end{bmatrix} + x \begin{bmatrix} 1 & 2a \\ 0 & 1 \end{bmatrix} + y \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 4a \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} x & 2ax \\ 0 & x \end{bmatrix} + \begin{bmatrix} y & 0 \\ 0 & y \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1+x+y & 4a+2ax \\ 0 & 1+x+y \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$4a+2ax=0$$

$$2ax = -4a$$

$$x = -2$$

$$1+x+y=0$$

$$1-2+y=0$$

$$y = 1$$

Ex. Let $A = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$. Find A^{100}

Solution

Past Exam

$$A^2 = AA = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} = A$$

$\therefore$

$$A^2 = A$$

$$A^3 = A^2 A = AA = A^2 = A$$

$\therefore$

$$A^3 = A$$

$$A^4 = A^3 A = AA = A^2 = A$$

$\therefore$

$$A^4 = A$$

$$\therefore A^{100} = A = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

$$ABA \neq A^2B \neq BA^2 \quad (62)$$

Let A, B are $n \times n$ matrices, such that $(A - B)^2 = 0$ and $A^2 = A$, $B^2 = B$

Past Exam

a) Show that $BA = A - AB + B$

b) Using part (a) to show that $(AB)^2 = AB$

$$(a) \quad (A - B)^2 = 0 \rightarrow (A - B)(A - B) = 0$$

$$A^2 - AB - BA + B^2 = 0$$

$$BA = A^2 - AB + B^2$$

|||

$$A^2 = A$$

$$B^2 = B$$

$$\boxed{BA = A - AB + B}$$

$$(b) \quad (AB)^2 = (AB)(AB) = A(BA)B$$

$$= A(A - AB + B)B$$

$$= (A^2 - A^2B + AB)B$$

$$= (A^2B - A^2B^2 + AB^2)$$

$$= AB - AB + AB = AB$$

|||

$$A^2 = A$$

$$B^2 = B$$

$$\therefore \boxed{(AB)^2 = AB}$$

Quiz

Let $A = \begin{bmatrix} 1 & 3 \\ 0 & -1 \end{bmatrix}$. Find A^{60}

Properties:

خواص عملية الضرب

If A , B and C are of the appropriate sizes

(1) $A(B \pm C) = AB \pm AC$

ننتبه للترتيب في الضرب

(2) $(B \pm C)A = BA \pm CA$

لا، لضرب أحدهما غير إبدالي

(3) $A(BC) = (AB)C \neq (AC)B$

عدد حقيقي

(4) $k(AB) = A(kB) = (kA)B$, $k \in R$

عامل مشترك

يبقى I

(5) $A^2 + 4A = A(A + 4I_n)$

(6) $A_n I_n = I_n A = A$

كل مصفوفة تضربها
بـ Identity تعطى
نفس المصفوفة بشرط
أن الضرب متاح

(7) $AB + A^2C = A(B + AC)$

مشتركة في الـ A

(8) $AB + CA \rightarrow$ لا يوجد مشترك بسبب

اختلاف موقع A [مرة يار ومرة يمين]

Transpose of a Matrix :

تدوير المصفوفة

$$A : m \times n \longrightarrow A^T : n \times m$$

Example: $A_{3 \times 1} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} \longrightarrow A^T_{1 \times 3} = [1 \quad 2 \quad 4]$

عند التدوير ... السطر يصير عمود .

Properties:

$$(1) (A^T)^T = A$$

$$(2) (A \pm B)^T = A^T \pm B^T$$

يتوزع على الجمع والطرح

$$(3) (rA)^T = r A^T, \quad r \in R$$

العدد لا يتأثر بالتدوير

$$(4) (AB)^T = B^T A^T$$

$$(ABC)^T = C^T B^T A^T$$

يتوزع ويعكس

$$(5) (I_n)^T = I_n, \quad (O_n)^T = O_n$$

Ex. Simplify $(3A^T - 2AB^T + 7I)^T$

$$= (3A^T)^T - (2AB^T)^T + (7I)^T$$

$$= 3(A^T)^T - 2(AB^T)^T + 7(I)^T$$

$$= 3A - 2BA^T + 7I$$

Find the matrix X satisfying**Past Exam**

$$\left(2X^T - 3 \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}\right)^T = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}.$$

$$2X - 3 \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}^T = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$$

$$2X - 3 \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$$

$$2X - \begin{bmatrix} 3 & -3 \\ 6 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$$

$$2X = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix} + \begin{bmatrix} 3 & -3 \\ 6 & 3 \end{bmatrix}$$

$$2X = \begin{bmatrix} 5 & 0 \\ 5 & 5 \end{bmatrix} \quad \div 2$$

$$\therefore X = \frac{1}{2} \begin{bmatrix} 5 & 0 \\ 5 & 5 \end{bmatrix}$$

$$X = \begin{bmatrix} \frac{5}{2} & 0 \\ \frac{5}{2} & \frac{5}{2} \end{bmatrix}$$

Past Exam

67

Let $B = \begin{bmatrix} 2 & 4 & 3 \\ -1 & 5 & 4 \end{bmatrix}$ & $Y = \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix}$.

If possible, find

(a) $(BY)Y^T + 2B$

(b) $(B^TY)Y^T + 2B^T$

$$(a) BY = \begin{bmatrix} 2 & 4 & 3 \\ -1 & 5 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

$2 \times 3 \quad 3 \times 1$

$$(BY)Y^T = \begin{bmatrix} 5 \\ 0 \end{bmatrix} \begin{bmatrix} 2 & -2 & 3 \end{bmatrix} = \begin{bmatrix} 10 & -10 & 15 \\ 0 & 0 & 0 \end{bmatrix}$$

$2 \times 1 \quad 1 \times 3$

$$(BY)Y^T + 2B = \begin{bmatrix} 10 & -10 & 15 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 4 & 8 & 6 \\ -2 & 10 & 8 \end{bmatrix}$$
$$= \begin{bmatrix} 14 & -2 & 21 \\ -2 & 10 & 8 \end{bmatrix}$$

$$(b) B^T = \begin{bmatrix} 2 & -1 \\ 4 & 5 \\ 3 & 4 \end{bmatrix}$$

$$B^TY = \begin{bmatrix} 2 & -1 \\ 4 & 5 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix} = \text{impossible}$$

$3 \times 2 \quad 3 \times 1$

[the number of columns of $B^T \neq$ the number of rows of Y]

$\therefore (B^TY)Y^T + 2B^T \rightarrow \text{Undefined}$

Remark: If AB is defined

① If one of A, B is zero matrix then $AB = 0$

② If A, B be two nonZero matrices then there are two cases

(i) $AB \neq 0$

for example

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 \\ -1 & 7 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & 7 \end{bmatrix} \\ = \begin{bmatrix} 1 & 2 \\ -1 & 7 \end{bmatrix}$$

(ii) $AB = 0$

for example

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \\ = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

③ If $A \neq 0$ then A^2 could be Zero matrix

for example

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$A^2 = AA = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Ex: Let A be 3×3 matrix such that
 $A = [a_{ij}] : a_{ij} = i - j$. Find $A + A^T$.

Text Book

$$A_{3 \times 3} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 0 & -1 & -2 \\ 1 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix}$$

$$A + A^T = \begin{bmatrix} 0 & -1 & -2 \\ 1 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & 1 \\ -2 & -1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O_3$$

Ex Let $A_{3 \times 3} = [a_{ij}] = \begin{cases} 0 & \text{if } i=j \\ i+j & \text{if } i > j \\ 2j & \text{if } i < j \end{cases}$
 Find A .

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 0 & 4 & 6 \\ 3 & 0 & 6 \\ 4 & 5 & 0 \end{bmatrix}$$

Linear Combination

70

Eng. Al-sfeer
99496600

$$\begin{aligned}\text{row}(AB) &= (\text{row } A) B \\ \text{col}(AB) &= A(\text{col } B)\end{aligned}$$

Ex. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 2 & 4 \\ 2 & 0 & 1 \end{bmatrix}$ $B = \begin{bmatrix} -1 & 0 & 4 \\ 2 & 6 & 2 \\ 1 & 1 & -1 \end{bmatrix}$

- (a) Express the third row of AB as a linear combination of the rows of B .
- (b) Express the second column of AB as a linear combination of the columns of A .

$$\begin{aligned}\text{(a) } \text{row}_3(AB) &= (\text{row}_3 A) B \\ &= (2) [-1 \ 0 \ 4] + (0) [2 \ 6 \ 2] + (1) [1 \ 1 \ -1]\end{aligned}$$

$$\text{(b) } \text{col}_2(AB) = A(\text{col}_2 B) = 0 \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} + 6 \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix}$$

(71)

Let $A = \begin{bmatrix} 3 & 2 & 0 \\ 5 & 1 & 8 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 1 \\ 4 & 2 \\ 2 & 7 \end{bmatrix}$

- (a) Express the first column of AB as a linear combination of columns of A .
- (b) Express the second row of AB as a linear combination of rows of B .

(a) $\text{col}_1(AB) = A(\text{col}_1 B)$

Past Exam

$$= 3 \begin{bmatrix} 3 \\ 5 \end{bmatrix} + 4 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 8 \end{bmatrix}$$

(b) $\text{row}_2(AB) = (\text{row}_2 A) B$

$$= 5 \begin{bmatrix} 3 & 1 \end{bmatrix} + 1 \begin{bmatrix} 4 & 2 \end{bmatrix} + 8 \begin{bmatrix} 2 & 7 \end{bmatrix}$$

Let

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & 3 \\ 5 & 0 & 1 \end{bmatrix} \text{ and } B = \begin{bmatrix} 2 & 0 & 1 \\ -1 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix}$$

72

(a) Find $\text{col}_2(AB)$

Text Book

$$\text{col}_2(AB) = A(\text{col}_2 B) =$$

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & 3 \\ 5 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 0 \end{bmatrix}$$

(b) Find $\text{row}_3(BA)$

$$\text{row}_3(BA) = (\text{row}_3 B) A$$

$$= [4 \ 0 \ 1] \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & 3 \\ 5 & 0 & 1 \end{bmatrix} = [9 \ 8 \ -3]$$

(c) Find $\text{col}_1(AA)$

$$\text{col}_1(AA) = A \text{col}_1(A) = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & 3 \\ 5 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 25 \\ 10 \end{bmatrix}$$

Let $A = \begin{pmatrix} 3 & 6 \\ -2 & 2 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & -3 \\ 1 & k \end{pmatrix}$.

- (a) Express the columns of AB as a linear combination of the columns of A . لم يحدد أي عمود
(b) Find all values of k for which $AB = BA$.

← لازم نأخذ كل عمود بدورح .

$$\text{Col}_1(AB) = 3 \begin{bmatrix} 3 \\ -2 \end{bmatrix} + 1 \begin{bmatrix} 6 \\ 2 \end{bmatrix} \quad \underline{\underline{1\text{-st column}}}$$

$$\text{Col}_2(AB) = -3 \begin{bmatrix} 3 \\ -2 \end{bmatrix} + k \begin{bmatrix} 6 \\ 2 \end{bmatrix} \quad \underline{\underline{2\text{-nd column}}}$$

لا مضا لم يتم تحديد أي عمود من B لذلك نأخذ كل عمود لعمود

$$(b) AB = BA \rightarrow \begin{bmatrix} 3 & 6 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} 3 & -3 \\ 1 & k \end{bmatrix} = \begin{bmatrix} 3 & -3 \\ 1 & k \end{bmatrix} \begin{bmatrix} 3 & 6 \\ -2 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 15 & -9+6k \\ -4 & 6+2k \end{bmatrix} = \begin{bmatrix} 15 & 12 \\ 3-2k & 6+2k \end{bmatrix}$$

$$-9+6k = 12 \rightarrow k = 21/6 = 7/2$$

$$-4 = 3-2k \rightarrow k = 7/2$$

The dot product of n -vectors

$$\text{Let } a = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} = (a_1, a_2, \dots, a_n)$$

$$\text{and } b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = (b_1, b_2, \dots, b_n) \text{ then}$$

$$a \cdot b = a_1 b_1 + a_2 b_2 + a_3 b_3 + \dots + a_n b_n$$

Ex: Find all values of r such that

$$u \cdot v = 0 \text{ where}$$

$$u = (2, r+1, -4) \text{ and } v = (2, r-2, r-1)$$

$$\therefore \vec{u} \cdot \vec{v} = (2, r+1, -4) \cdot (2, r-2, r-1) = 0$$

$$4 + (r+1)(r-2) + (-4)(r-1) = 0$$

$$4 + r^2 - 2r + r - 2 - 4r + 4 = 0$$

$$r^2 - 5r + 6 = 0 \rightarrow (r-2)(r-3) = 0$$

$$\boxed{r=2}$$

$$\boxed{r=3}$$

Quiz

75

Past Exam

(20 pts) Let $A = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 & 2 \\ 3 & -2 & 1 \end{bmatrix}$. If possible, compute:

- (a) A^2 (b) $A - 2B$ (c) $B^T B$ (d) $\text{Row}_1(A) \cdot \text{Col}_1(B)$.

$$(a) A^2 = AA = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 5 & 1 \\ 1 & 2 \end{bmatrix}$$

$$(b) A - 2B = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} - \begin{bmatrix} 0 & 2 & 4 \\ 6 & -4 & 2 \end{bmatrix} = \text{NOT defined}$$

because they have different size

$$(c) B^T B = \begin{bmatrix} 0 & 3 \\ 1 & -2 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 \\ 3 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 9 & -6 & 3 \\ -6 & 5 & 0 \\ 3 & 0 & 5 \end{bmatrix}$$

$3 \times 2 \quad \quad 2 \times 3 \quad \quad \quad 3 \times 3$

$$(d) \text{Row}_1(A) \cdot \text{Col}_1(B) = [2 \ 1] \cdot \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$
$$= 0 + 3 = 3$$

Trace of a Matrix $[tr(A)]$

is defined to be the sum of the entries on the main diagonal of A .

$$tr(A) = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + a_{33} + \dots + a_{ii}$$

Remark:

If A is NOT a square matrix then $tr(A)$ is undefined.

Eng. Al-sfeer
99496600

Ex:

$$(1) A = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix} \rightarrow tr(A) = 1 + 2 = 3$$

$$(2) B = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 4 & 1 \\ 0 & -1 & 3 \end{bmatrix} \rightarrow tr(B) = 1 + 4 + 3 = 8$$

$$(3) C = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 1 & 5 \end{bmatrix} \rightarrow tr(C) \text{ is undefined}$$

because C is NOT a square matrix.

Properties of Trace of a Matrix

$$(1) \text{tr}(A \pm B) = \text{tr}(A) \pm \text{tr}(B) \quad \text{يتوزع على الجمع والطرح}$$

$$(2) \text{tr}(cA) = c \text{tr}(A) \quad ; \quad c \in \mathbb{R}$$

$$(3) \text{tr}(AB) = \text{tr}(BA)$$

$$(4) \text{tr}(A) = \text{tr}(A^T) \quad T \rightarrow \text{لا يؤثر}$$

$$(5) \text{tr}(I_n) = n \quad \text{tr}(O_n) = 0$$

$$(6) \text{tr}(AB) \neq \text{tr}(A) \text{tr}(B) \quad \text{in general}$$

$$\text{Sometimes } \text{tr}(AB) = \text{tr}(A) \text{tr}(B)$$

$$(7) \text{tr}(A^n) \neq (\text{tr} A)^n \quad \text{in general}$$

Let $A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$

Text Book

(a) Find $\text{tr}(A^2)$

$$A^2 = AA = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} -1 & -4 \\ 8 & 7 \end{bmatrix}$$

$$\text{tr}(A^2) = -1 + 7 = 6$$

(b) Find $\text{tr}(2A^T - 3I_2)$

$$= \text{tr}(2A^T) - \text{tr}(3I_2)$$

$$= 2\text{tr}(A^T) - 3\text{tr}(I_2)$$

$$= 2\text{tr}(A) - 3(2)$$

$$= 2(4) - 6 = 8 - 6 = 2$$

(c) Find $\text{tr}(AA^T)$

$$AA^T = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 13 \end{bmatrix}$$

$$\text{tr}(AA^T) = 2 + 13 = 15$$

Q9

Let $A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ 0 & 1 \\ 1 & 3 \end{bmatrix}$. If possible compute

Past Exam

- (a) $A + B^T$ (b) $B^T A$ (c) $2(A + A^T)$

(a) $A + B^T$ is NOT possible because
 3×3 2×3
 Size of $A \neq$ size of B^T

$$(b) B^T A = \begin{bmatrix} 1 & 0 & 1 \\ -2 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 4 & -4 \end{bmatrix}$$

2×3 3×3 2×3

$$(c) 2(A + A^T) = 2 \left(\begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 1 & 1 & -1 \end{bmatrix} + \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & -1 \end{bmatrix} \right)$$

$$= 2 \begin{bmatrix} 2 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & -2 \end{bmatrix} = \begin{bmatrix} 4 & 4 & 4 \\ 4 & 4 & 4 \\ 4 & 4 & -4 \end{bmatrix}$$

Consider $A = \begin{bmatrix} 3 & 2 \\ 1 & a \end{bmatrix}$. Find the values of (a) such that A is row equivalent to I_2 .

$$A \xrightarrow[\text{row equivalent}]{\text{RREF}} I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 2 \\ 1 & a \end{bmatrix} \quad R_1 \leftrightarrow R_2$$

$$\begin{bmatrix} 1 & a \\ 3 & 2 \end{bmatrix} \xrightarrow{-3R_1 + R_2 \rightarrow R_2}$$

$$\begin{bmatrix} 1 & a \\ 0 & 2-3a \end{bmatrix} \rightarrow \begin{matrix} \text{leading} \\ \text{المفروض ما ياتي صفر} \end{matrix}$$

$$2-3a \neq 0 \rightarrow a \neq \frac{2}{3}$$

$$a \in \mathbb{R} \setminus \left\{ \frac{2}{3} \right\}$$

- 42- Find a 3×1 matrix X with entries not all zero such that $AX = 3X$ where

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & -4 & 5 \end{bmatrix}$$

إذا لم نأخذ
X عامل مشترك
نفس
Homogeneous system.

$$\because AX = 3X \rightarrow AX - 3X = 0$$

$$(A - 3I_3)X = 0 \rightarrow \text{Homogeneous system}$$

$$\text{Let } B = A - 3I_3 = \begin{bmatrix} 1 & 2 & -1 \\ 1 & 0 & 1 \\ 4 & -4 & 5 \end{bmatrix} - 3 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} -2 & 2 & -1 \\ 1 & -3 & 1 \\ 4 & -4 & 2 \end{bmatrix}$$

$$[B | 0] = \left[\begin{array}{ccc|c} -2 & 2 & -1 & 0 \\ 1 & -3 & 1 & 0 \\ 4 & -4 & 2 & 0 \end{array} \right] \quad R_1 \leftrightarrow R_2$$

$$\left[\begin{array}{ccc|c} 1 & -3 & 1 & 0 \\ -2 & 2 & -1 & 0 \\ 4 & -4 & 2 & 0 \end{array} \right] \xrightarrow{\substack{2R_1 + R_2 \rightarrow R_2 \\ -4R_1 + R_3 \rightarrow R_3}} \left[\begin{array}{ccc|c} 1 & -3 & 1 & 0 \\ 0 & -4 & 1 & 0 \\ 0 & 8 & -2 & 0 \end{array} \right] \xrightarrow{-\frac{1}{4}R_2 \rightarrow R_2}$$

$$\left[\begin{array}{ccc|c} 1 & -3 & 1 & 0 \\ 0 & 1 & -1/4 & 0 \\ 0 & 8 & -2 & 0 \end{array} \right] \xrightarrow{\substack{3R_2 + R_1 \rightarrow R_1 \\ -8R_2 + R_3 \rightarrow R_3}} \left[\begin{array}{ccc|c} 1 & 0 & 1/4 & 0 \\ 0 & 1 & -1/4 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1/4 t \\ 1/4 t \\ t \end{bmatrix} = t \begin{bmatrix} -1/4 \\ 1/4 \\ 1 \end{bmatrix} \quad \text{nontrivial solution}$$